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Methodology for Commercial Vehicle Mechanical Systems Maintenance: Data-Driven and Deep-Learning-Based Prediction

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ABSTRACT This paper presents a predictive maintenance (PdM) strategy for commercial vehicles, focusing on the turbocharger—a critical yet often under-monitored component. By combining sensor signals, workshop maintenance logs, and technical specifications, the study demonstrates how data-driven deep-learning techniques can robustly identify pending failures. Specifically, Long Short-Term Memory (LSTM) and Bidirectional LSTM (BiLSTM) architectures were employed to capture temporal dependencies and detect patterns that conventional approaches and purely onboard monitoring might overlook. Results on real-world fleet data indicate that BiLSTM achieved higher recall (98.65%) and a lower cost-score than standard LSTM, highlighting its effectiveness in minimizing missed failures. Although BiLSTM incurred slightly higher computational overhead, its superior performance underscores the value of integrating multi-sourced data and advanced sequence models for reliable, actionable PdM in heavy-duty fleets.

INDEX TERMS Commercial Vehicles Industry, Data Analysis, Data-Driven, Deep Learning, Predictive Maintenance of Mechanical Systems.

I. INTRODUCTION

THE transportation industry is undergoing a profound transformation driven by emerging technologies and new business models, such as vehicle connectivity, shared mobility services, on-demand platforms, and the pursuit of sustainable solutions like electric vehicles. These shifts create a highly disruptive landscape, compelling companies to adopt innovative strategies to remain competitive. As a result, investments in data analytics and artificial intelligence (AI) have become essential for both market success and corporate sustainability [1], [2].

Commercial vehicles are pivotal in modern supply chains, particularly in regions that rely heavily on road transport infrastructure. Robustness, high reliability, and minimal downtime are key factors for maintaining profitability and competitiveness in these fleets [3]. However, increasing technological complexity—multiple mechanical subsystems, diverse sensors, and intricate onboard electronics—leads to higher probabilities of failures. In practice, unplanned vehicle stoppages can cause significant financial losses, delivery delays,

and degrade customer service quality [4]. Thus, minimizing unexpected downtime is paramount to preserving operational capacity and reducing logistics costs.

Predictive maintenance (PdM) has gained traction in response to these challenges as an effective methodology to anticipate potential failures and avoid unscheduled stops [5]. When combined with data-driven models and deep learning techniques, PdM can achieve high Overall Equipment Effectiveness (OEE) (above 90% in some instances) and yield significant returns on investment [6], [7]. Additionally, transitioning from rigid, mileage-based maintenance schedules to more flexible predictive routines can decrease downtime, optimize spare parts inventory, and improve overall fleet management [2].

However, successfully implementing data-driven and deep-learning-based PdM in heavy-duty vehicles entails multiple hurdles. A key difficulty is identifying the root causes of failures, as Diagnostic Trouble Codes (DTCs) are often triggered when the subsystem performs below predefined thresholds or has already failed [8]. Moreover, limited onboard sensors

and insufficient coverage of all failure modes complicate real-time monitoring [9]. Consequently, relying solely on onboard detection can be inadequate.

This work proposes an enhanced predictive methodology that exploits operational sensor data and incorporates additional information collected in specialized workshops. By unifying data from fault identification, repair logs, maintenance history, and product specifications, we can comprehensively link patterns in the data with mechanical degradation processes through deep learning.

As a case study, we examine the turbocharger of a commercial vehicle. Turbochargers often lack dedicated condition-monitoring sensors; hence, malfunctions are typically detected only when drivers perceive performance issues. Although specific engine and after-treatment sensors can reflect turbocharger performance, these signals alone may not sufficiently capture early warning signs. Workshops equipped with advanced diagnostic tools can further detect factors related to turbocharger failures, yet such data are unavailable to drivers during daily operations. Therefore, leveraging multiple data sources—from onboard telemetry to off-board workshop logs—can be highly advantageous for timely fault detection.

In short, we aim to demonstrate the effectiveness of our deep-learning-based predictive maintenance approach for commercial vehicles, showcasing how the fusion of specification data (TSD), maintenance records (MD), and operational data (OD) augments failure prediction. The main contributions are:

- A novel PdM strategy for heavy-duty trucks, integrating technical specifications, workshop maintenance logs, and operational signals to address gaps in sensor coverage.
- A large-scale data-driven framework that employs deep learning models, validated with real-world commercial-vehicle data, to enhance repair efficiency and standardize onboard and off-board data collection.
- An implementation of Long Short-Term Memory (LSTM) and Bi-Directional LSTM (BiLSTM) architectures for diagnosing and predicting turbocharger faults, improving fleet availability.
- A comparative analysis of LSTM vs. BiLSTM performance in predicting turbocharger failures, highlighting their advantages.

The paper is organized as follows. Section II outlines recent studies on data-driven deep-learning methods for predictive maintenance and provides an overview of turbocharger functionality. Section III describes our methodology, including data processing and deep-learning architectures. Section IV presents the experimental results. Finally, Section V concludes the work and discusses potential future directions.

II. RELATED WORK

In recent years, deep learning has emerged as a major cornerstone in predictive maintenance (PdM) for commercial vehicles, motivated by the increasing availability of sensor

data and the need for automated fault detection [10], [11]. Broadly, PdM methodologies can be divided into two categories: *onboard* approaches, which analyze sensor data in real time, and *off-board* approaches, which leverage greater computational power and extensive historical logs [12]. Depending on the specific goal, these methods may focus on predicting components' Remaining Useful Life (RUL), detecting anomalies in multivariate time series, or classifying failures using supervised learning.

A longstanding challenge in PdM research is thorough data handling—covering data cleaning, feature selection, and class balancing. For example, [13] employed vibration signals of spark-ignition engines to diagnose valve-system defects, extracting statistical features of reduced dimensionality to identify mechanical anomalies efficiently. Likewise, [14] applied Random Forest models and filter-based feature selection to predict failures in truck air compressors, mitigating class imbalance with the Synthetic Minority Oversampling Technique (SMOTE) [15]. More recently, efforts have expanded to Generative Adversarial Network (GAN) architectures, wherein [16] demonstrated how GAN variants can generate synthetic samples of rare failure modes, improving overall classification performance.

Deep learning architectures that model sequential dependencies—such as Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM), and Bidirectional LSTM (BiLSTM)—have also gained traction. A systematic survey by [17] reported that Random Forest, classical neural networks, and RNN-based approaches remain common in industrial environments, providing powerful tools for time-series analysis. For instance, [18] highlighted that LSTM networks can exhibit higher stability than traditional machine-learning models when forecasting compressor failures over extended horizons. Meanwhile, [19] introduced a holistic data-lifecycle strategy that considerably reduced downtime by integrating medium- and long-term predictions into maintenance workflows.

In parallel, novel architectures like Transformers have been proposed to capture long-range dependencies in predictive maintenance. For example, [20] illustrated how attention-based mechanisms outperform conventional RNNs in detecting anomalies in industrial sensor data, leveraging multi-head attention to emphasize critical temporal segments. Similarly, [21] combined Transformer encoders with graph neural networks for multi-sensor fusion, enabling real-time fault prediction in complex manufacturing pipelines. Another emerging trend is federated learning, wherein multiple organizations train models collaboratively without sharing raw data; [22] developed a framework to preserve data privacy across commercial fleets while still achieving effective PdM outcomes.

Despite these advances, many studies assume comprehensive sensor coverage or specialized instrumentation for each target subsystem. Critical components like turbochargers often lack dedicated sensing in sizeable commercial vehicle fleets. Consequently, merging data from product specifications, historical maintenance logs, and limited onboard met-

rics proves crucial for early detection of subsystem degradation. Even though state-of-the-art techniques like Transformers have shown promise in modeling high-dimensional time series [23], [24], LSTM and BiLSTM architectures remain particularly compelling in contexts where sensor data is partially available or training samples are relatively limited.

Although the works above offer valuable insights, many focus on a single data domain (e.g., exclusively onboard sensor data) or assume extensive instrumentation. By contrast, our approach integrates data from multiple sources (technical specifications, workshop records, and partial telematics), thus addressing scenarios where comprehensive sensing is not feasible. For instance, while CNN or Transformer-based frameworks may excel when large and high-frequency signals are continuously recorded, LSTM/BiLSTM proves more suitable for moderate data volumes and incomplete sensor coverage—a finding that aligns with initial tests, where CNN-based approaches underperformed in such conditions. This interplay of lesser sensor availability and the need for stable sequence modeling underpins our motivation to adopt LSTM/BiLSTM.

Finally, to address the persistent issues of data imbalance and missing values—points often underemphasized in the existing literature—we incorporate SMOTE to balance rare fault classes and carefully designed imputation methods for incomplete signals. Authors like [13] and [14] have already shown how preprocessing steps can strongly influence results, while [16] has demonstrated that synthetic oversampling can preserve classifier integrity. Building on their conclusions, we adapt these techniques to our scenario, acknowledging that imputation and SMOTE require careful tuning to avoid unintended distortion in real-time performance.

In essence, this study complements prior research by uniting partial sensor coverage with additional sources of information, such as product specifications and workshop logs, and by applying RNN-based models that remain robust under constrained data conditions. In subsequent sections, we explain how our methodological choices—from architectural design to data preprocessing—yield a scalable, reproducible, and more thorough predictive maintenance solution for commercial vehicles, aiming to fill the gap left by purely onboard or single-dataset investigations.

A. THE TURBOCOMPRESSOR FUNCTIONING, SIGNAL COLLECTING, PROCESSING, AND MONITORING

Alfred Büchi invented the turbocharger in 1905. It is an improvement for internal combustion engines that helps increase the air density admitted to the combustion chamber. Increasing the intake air density improves power output and efficiency compared to a naturally aspirated engine.

The turbocharger is considered a mechanical system with a stable and long-lasting design, which does not undergo many variations between generations of vehicles. It also presents gradual wear patterns that develop over time. Due to these and other characteristics, it is an ideal subject for predictive maintenance.

The turbine of an engine is powered by the exhaust gases captured in the casing and directed to the rotor. This process provides work for the centrifugal compressor, increasing the air intake into the cylinder. The system uses energy that would otherwise be wasted in exhaust gas pressure and temperature, optimizing the engine's efficiency. The relief valve reduces the flow in the turbine to prevent the turbocharger from exceeding its maximum speed. It ensures the system does not surpass the specified pressure limit. Fig. 1 visually represents the turbocharger's operating cycle.

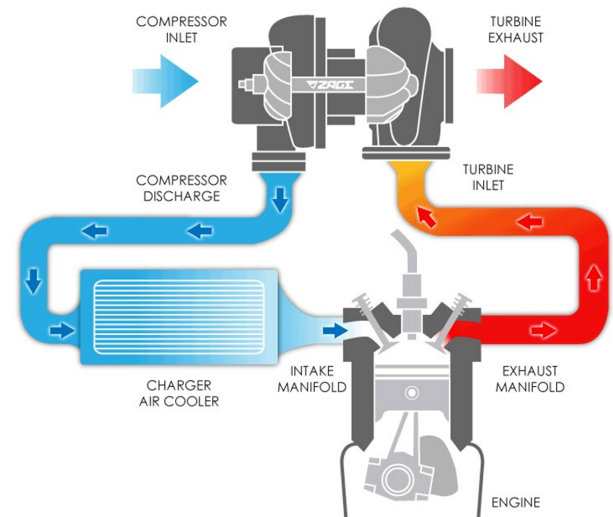


FIGURE 1. Turbocharger cycle.

The engine cylinder's exhaust gases are directed to the turbine before being discarded. The turbine's rotor blades extract energy from the exhaust gases and transform it into shaft power to rotate the engine compressor. As the compressor rotor blades rotate, they compress the air and increase pressure. However, the compressed air also increases in temperature, causing a decrease in its density. To combat this effect, the hot compressed air passes through a heat exchanger to reduce its temperature and increase density. Finally, the denser compressed air is admitted into the engine cylinder.

Turbochargers are critical to a truck's operations, and their failure can result in unplanned vehicle downtime and increased service costs for the maintenance provider. The turbocharger's condition is primarily monitored by recording internal and environmental operating parameters, as it does not have a dedicated sensor to monitor its condition. However, data from engine sensors and the after-treatment system can help identify the turbocharger's operating condition.

Important factors related to turbocharger failures include exhaust gas temperature (before and after turbocharging), air restriction and quality, pressure, engine speed and load, oil temperature, fuel consumption, and engine oil consumption.

The automotive industry has made significant efforts to implement self-diagnostics for vehicles' mechanical systems to enable fault detection and create efficient methods to obtain

data from Electronic Control Units (ECUs) for data-based fault analysis. The ECU can monitor input and output signals and manage its communication with a vehicle's components and other external networks.

Monitoring and analyzing the data or variables generated by the vehicle ECUs (Electronic Control Units) enables the detection of anomalies or faults in various vehicle components, including the engine and brakes. An engine ECU can provide over 20,000 variables accessible through a direct Controller Area Network (CAN) connection using a transmission protocol such as the CAN Calibration Protocol (CCP) or the Universal Measurement and Calibration Protocol (XCP). However, due to bandwidth limitations, only a small subset of these variables can be collected (typically between 100 and 130, depending on the sample rate), although the volume of collected data is still substantial [25]. Fig. 2 illustrates an ECU with inputs from various sensors, outputs sent to different actuators, and monitoring functions implemented directly in the hardware.

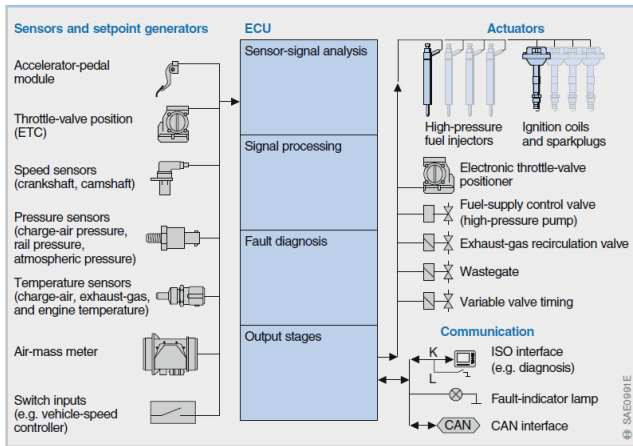


FIGURE 2. ECU inputs, outputs, signal processing, and onboard diagnosis modules. Source: [26]

Among other functions performed by the ECU, filter application is commonly used to alter signal characteristics concerning frequency and phase. A filter extracts critical frequencies in a signal by attenuating unwanted frequencies. In a standard ECU, numerous filters with different features rely on the frequency domain response, such as low pass filters, high pass filters, band pass filters, moving average filters, and band stop filters [27]. The deployment of filters is fundamental to preventing signal corruption by noise and interference, thereby preserving signal integrity and making the data available either for controlling actuators or for extracting the processed variables via the vehicle CAN.

III. METHODOLOGY

This section outlines the methodology. It consists of five stages applied systematically to ensure correct development of the proposed method and implementation of deep learning algorithms. Section III-A explains the approach used to understand the turbocharger problem, which can be applied to

any mechanical system of commercial vehicles. Section III-B describes how data are collected from various sources and used to compose a training data set. Section III-C explains the methodology for data preparation, including cleaning, balancing analysis, and normalization of the collected data. Section III-D discusses the selected deep learning algorithms and their implementation. Finally, Section III-E explains how the proposed deep learning models are trained, evaluated, and validated.

A. PROBLEM UNDERSTANDING

This research aims to develop and implement a model using deep learning algorithms to predict the integrity of systems in commercial vehicles. The model uses Technical Specification Data (TSD), Operational Data (OD), Fault Codes (FC), and Maintenance Data (MD).

We employ the sliding-box approach to address the prediction problem, a binary classification task commonly used in predictive maintenance studies. In this approach, the model predicts whether a component will fail within a certain period based on data collected up to a specific point.

A critical aspect of our methodology is the choice of deep learning architectures. While Convolutional Neural Networks (CNNs) and Transformers have demonstrated significant capabilities in handling complex patterns and long-range dependencies in time-series data, they present certain limitations in our context. CNNs excel at extracting localized temporal features but may struggle with capturing long-term dependencies unless augmented by additional layers or large training data. Transformers, in turn, can be computationally intensive and prone to overfitting with smaller datasets. In contrast, Long Short-Term Memory (LSTM) and Bidirectional LSTM (BiLSTM) networks offer a balanced solution by effectively capturing essential sequence dependencies while maintaining computational efficiency, making them particularly suitable for moderate-sized datasets and partial sensor coverage.

Initial experiments demonstrated that LSTM/BiLSTM models outperform CNN-based approaches in scenarios with limited sensor data and moderate training sample sizes. These findings are supported by the literature, which indicates that LSTM/BiLSTM networks provide higher stability and robustness in predictive maintenance tasks where sensor coverage is incomplete or data is imbalanced.

Leveraging LSTM/BiLSTM architectures offers several key benefits in predicting turbocharger failures. These models handle the temporal dependencies inherent in turbocharger performance and degradation, enabling the detection of gradual wear patterns and early warning signs preceding failures. Moreover, in commercial vehicles where not all components are equipped with dedicated sensors, LSTM/BiLSTM can integrate data from multiple sources, including operational metrics and maintenance logs, compensating for gaps in sensor coverage. This multi-source data fusion enhances the model's ability to predict failures even with incomplete real-time data.

Fig. 3 depicts the Sliding-Box method, where the classifier indicates whether the component will fail within a particular time window Δt . Each horizontal line denotes the timeline of a vehicle, and the events can include readings of operational data and fault codes, among others.

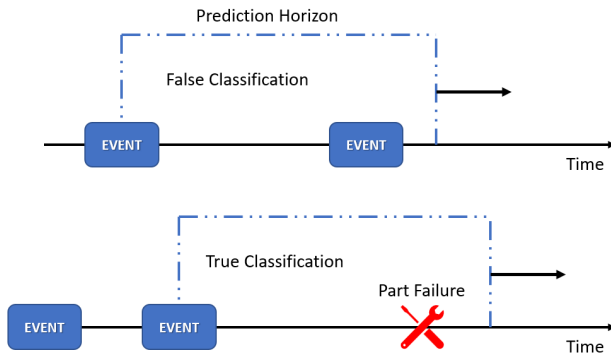


FIGURE 3. Sliding-Box binary classification model.

We obtain samples of defective commercial vehicles belonging to the positive class by establishing reading records in the operational data set within a Δt period, covering 90 days of records gathered before restoring or substituting the turbocharger component. All other commercial vehicle registrations are considered to belong to the negative class. For healthy commercial vehicles without repair or component replacement, all available records are selected and classified as belonging to the negative class. Fig. 4 briefly illustrates the proposed category extraction.

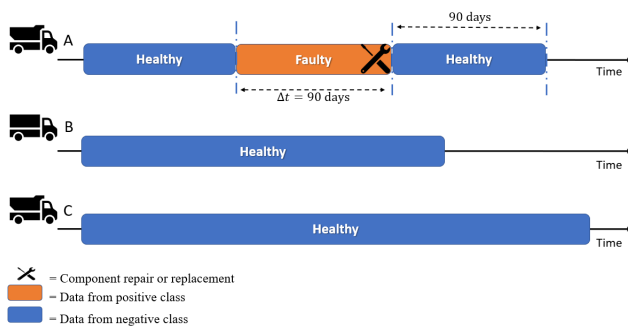


FIGURE 4. Strategy for extracting categories.

Thus, compared to other models, our approach using LSTM/BiLSTM within the sliding-box framework effectively addresses the challenges posed by temporal dependency modeling and partial sensor coverage. Unlike CNNs, which may require extensive feature engineering and larger datasets to capture longer-range temporal dependencies, or Transformers, which demand significant computational resources and risk overfitting with limited data, LSTM/BiLSTM networks inherently manage temporal sequences and exhibit strong robustness under moderate data conditions. This makes them a pragmatic choice for a predictive maintenance framework, ensuring both reliability and

efficiency in predicting turbocharger failures in commercial vehicles.

B. DATA UNDERSTANDING

The data is sourced from a centralized repository called the Data Lake, which aggregates information from various origins within the commercial vehicle industry. The Data Lake employs a storage layer managed by Hadoop, a framework designed for efficiently processing and storing large datasets. Additionally, it incorporates a query layer facilitated by tools such as Spark, Hive, and HBase, enabling seamless data access and manipulation.

Operational data is collected from multiple vehicle systems through sensors that transmit signals to the vehicle's Electronic Control Module (ECM). These signals are processed and transformed into meaningful physical quantities, aiding in monitoring vehicle performance. Communication between ECUs is managed via the Controller Area Network (CAN), which is pivotal in controlling vehicle operations and assessing the operational condition.

Vehicles have communication modules that receive real-time data and transmit it to the data storage center through GSM networks. Furthermore, data is downloaded and stored in the system whenever a vehicle undergoes maintenance or other activities at a workshop. This capability is crucial for developing various applications and systems leveraging the collected data for predictive maintenance.

Automotive companies accumulate extensive data over years of operation, encompassing project drawings, technical reports, maintenance and failure histories, and vehicle operation statistics. This diverse data is managed using specialized software and applications, all stored within the central Data Lake repository. The comprehensive nature of this data serves as an invaluable resource for designing predictive models to enhance the reliability and efficiency of mechanical systems in commercial vehicles.

To ensure a consistent dataset and reflect real operational environments, a specific population of commercial vehicle models and applications is defined. The scope is restricted to vehicles equipped with Euro V and equivalent engines manufactured from 2018 onwards. Additionally, the selection criteria focus on vehicles operating in Europe and South America, chosen for their technical similarity and comparable climatic conditions.

Vehicles are prioritized based on the availability and quality of their readings to ensure a robust dataset. Events within the dataset are labeled as either failures (denoted by 1) or non-failures (denoted by 0) based on the occurrence of part repairs or replacements corresponding to specific dates and mileages. This labeling is fundamental for training the predictive models to distinguish between healthy and failing components.

Post-processing reorganizes the data structure so that each column represents a specific parameter of interest. This results in time-disaggregated data that captures the vehicle's damage patterns and facilitates failure prediction. This struc-

tured approach supports detailed data analysis, extracting meaningful insights and trends.

Exploratory data analysis and visualization are critical for understanding the dataset's underlying patterns. Using Python's libraries—such as NumPy, Pandas, and Matplotlib—statistical analyses and visualizations assess data integrity and uncover relationships between variables. Data quality is meticulously inspected by evaluating the presence of null values, ensuring the reliability of the dataset. Missing values are addressed through appropriate imputation techniques, while class imbalances are identified and mitigated to guarantee a balanced representation in the predictive models.

Furthermore, the relationships between feature variables are analyzed to detect high correlations that could lead to overfitting. Techniques such as dimensionality reduction and feature selection minimize multicollinearity, enhancing the model's generalization capabilities. The NumPy library supports mathematical computations and matrix operations, ensuring the data is well-prepared for subsequent modeling stages.

C. DATA PREPARATION

Operational data is collected either when the vehicle visits a workshop or is transmitted to the Data Lake via 4G or 5G networks. However, data is not always collected at regular intervals, posing significant challenges for deep learning models, which typically require uniformly sampled time series data. To address this, the data is categorized according to time and distance traveled and subsequently grouped into bands of 5,000 kilometers. This grouping ensures that each data point represents a meaningful interval of vehicle operation, enhancing the temporal resolution of the dataset. When no readings are available within a specific band, the preceding reading fills the gap, resulting in a continuous time series for each commercial vehicle. This imputation strategy maintains the temporal continuity necessary for deep learning models to effectively capture temporal dependencies.

This technique enables the use of deep learning models by providing a comprehensive view of past and future operational states, maximizing the information available for learning. By ensuring that the time series data is uniformly sampled, the models can better identify patterns and trends indicative of component failures, such as those of the turbocharger.

To further enhance data quality, variables with over 65% null values are removed from the dataset. This threshold was selected based on preliminary analyses indicating that variables exceeding this level of missingness contribute minimal value to the predictive models and may introduce noise or bias. By excluding these highly incomplete variables, we reduce the dimensionality of the dataset and focus on features that provide reliable and meaningful information for failure prediction.

For the remaining variables with less than 65% missing values, we employ Mean Imputation (MI) to substitute the missing entries. MI estimates the average value for each

feature based on the available training data and replaces null values accordingly. This method was chosen for its simplicity and effectiveness in maintaining the overall data distribution without introducing significant bias. Applying MI consistently to the training and validation datasets ensures uniform data preprocessing and stability in the deep learning models.

Addressing class imbalance, a common challenge in predictive maintenance datasets, is also crucial. We employ the Synthetic Minority Over-sampling Technique (SMOTE) from the imbalanced-learn library in Python. SMOTE generates synthetic samples for the minority class by interpolating between existing instances, thereby enhancing the representation of failure events in the dataset [15]. This augmentation reduces classification bias and allows the models to learn the characteristics of both classes more effectively, leading to accurate and reliable failure predictions.

Normalization is performed to harmonize the scale of all features, which is vital for training deep learning models efficiently. We apply the MinMaxScaler algorithm from the Scikit-learn library to scale each feature's values to the range [0, 1]. This normalization ensures that no feature disproportionately influences the model's learning process, enhancing training efficiency and accuracy. By preserving the original data distribution within the scaled range, we maintain the intrinsic relationships between features.

D. MODELING DEEP LEARNING MODELS

Recurrent Neural Networks (RNNs), particularly Long Short-Term Memory (LSTM) networks, are widely used for time-dependent forecasting due to their ability to capture long-term dependencies while mitigating gradient-related issues [28]. LSTM networks consist of cell blocks featuring three gates: the input gate, the forget gate, and the output gate. These gates regulate the flow of information into and out of the cell, allowing the network to retain relevant information over extended periods. This capability makes LSTM models well-suited for predicting component failures in commercial vehicles, where temporal dependencies play a crucial role.

Fig. 5 illustrates the structure of an LSTM cell:

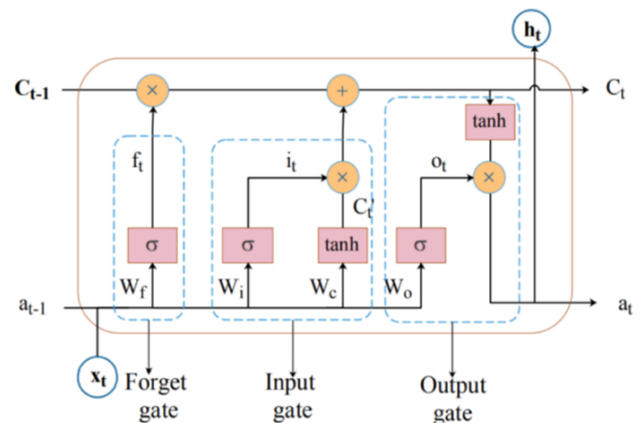


FIGURE 5. Basic diagram of an LSTM cell.

Inputs to the LSTM cell consist of the current input vector x_t and the previous hidden state $\alpha_{(t-1)}$. These inputs are processed through the three gates—input, forget, and output—each using sigmoid activation functions σ to determine how much information to retain or discard. The cell and hidden states employ hyperbolic tangent functions ($\tanh$) to introduce nonlinearity. The weight matrices (W_f, W_i, W_c, W_o) and bias vectors (b_f, b_i, b_c, b_o) are learned parameters that facilitate the gating mechanisms.

Forget Gate:

$$f_t = \sigma(W_f \times [\alpha_{(t-1)}, x_t] + b_f). \quad (1)$$

This gate decides how much of the previous cell state C_{t-1} should be retained.

Input Gate:

$$i_t = \sigma(W_i \times [\alpha_{(t-1)}, x_t] + b_i), \quad (2)$$

$$C'_t = \tanh(W_c \times [\alpha_{(t-1)}, x_t] + b_c). \quad (3)$$

These equations govern the incorporation of new information into the cell state.

$$C_t = f_t C_{t-1} + C'_t i_t. \quad (4)$$

Output Gate:

$$o_t = \sigma(W_o \times [\alpha_{(t-1)}, x_t] + b_o), \quad (5)$$

$$a_t = h_t = o_t \times \tanh(C_t). \quad (6)$$

This gate determines the hidden state h_t and the output a_t .

While traditional RNNs can learn temporal dependencies, they often struggle to capture long-term dependencies due to issues like vanishing gradients. LSTM networks address this limitation by incorporating gating mechanisms that regulate the flow of information, enabling the network to retain relevant information over longer sequences.

For certain maintenance tasks, it is crucial to consider both past and future information to improve prediction accuracy. This consideration motivates the use of Bidirectional LSTM (BiLSTM) networks. BiLSTM networks consist of two LSTM layers: one processes the sequence in the forward direction, while the other processes it in the backward direction. This bidirectional processing allows the network to capture information from both past and future contexts, providing a more comprehensive understanding of the data [29].

Architecture Specifications: Our model architecture consists of two stacked LSTM layers followed by a Bidirectional LSTM layer to capture temporal dependencies from both directions. Specifically, the first LSTM layer has 128 hidden units, and the second has 64. The Bidirectional LSTM layer is configured with 32 hidden units in each direction, resulting in 64 units total. Dropout layers with a rate of 0.3 are interspersed between the LSTM layers to prevent overfitting by randomly deactivating a subset of neurons during training.

The final layer is a Dense layer with a sigmoid activation function, suitable for binary classification tasks.

Design Choices Justification: The complexity and size of the dataset guided the selection of the number of layers and hidden units. With over 2.3 million observations and more than 100 features, the dataset requires a model capable of learning intricate patterns without becoming excessively large. The initial LSTM layers with 128 and 64 hidden units are designed to capture high-level temporal features, while the Bidirectional LSTM layer further improves the model's ability to understand dependencies from both past and future contexts.

Implementation Details: We use the Keras Sequential API to construct the LSTM and BiLSTM layers. Each LSTM layer is followed by a Dropout layer to provide regularization. The Dense layer at the end uses a sigmoid activation function to compress the output to a range between 0 and 1. The model is compiled with binary cross-entropy loss, which is suitable for binary classification tasks, and the Adam optimizer, known for its efficiency.

For the BiLSTM network, the LSTM layers are replaced with Bidirectional LSTM layers to enable the forward and backward processing of input data. This addition allows the model to leverage information from the entire sequence, thereby improving prediction accuracy. Hyperparameters, including the number of layers, hidden units, dropout rates, and learning rates, were optimized using five-fold cross-validation [30] to ensure the model's robustness.

E. TRAINING, TESTING, AND EVALUATING

The dataset is divided into two sets: the training set comprising 80% of the data and the testing set containing the remaining 20%. Random sampling is applied to select training and validation data for each model. With Scikit-learn's train-test split function, the dataset is partitioned while maintaining the class distribution in both sets, crucial for unbiased model evaluation.

We implement a five-fold cross-validation strategy to reduce overfitting and enhance the model's generalization. The training dataset is divided into five equally sized folds, where four folds are used for training and one fold is reserved for validation in each iteration. This approach ensures that the model is trained and validated on different subsets of the data, providing a comprehensive assessment of its performance across various data distributions. The choice of five folds balances computational efficiency and reliable performance estimates.

During training, hyperparameter tuning is performed to optimize the model's architecture and learning parameters. Key hyperparameters, such as the number of LSTM/BiLSTM layers, the number of hidden units per layer, dropout rates, learning rates, and batch sizes, are systematically varied within predefined ranges. Grid search combined with cross-validation is utilized to identify the optimal combination of hyperparameters that yields the highest validation performance. Selection criteria are based on maximizing the Area

Under the ROC Curve (AUC) while ensuring computational feasibility and preventing overfitting.

After cross-validation, the average model performance is computed based on the results obtained from each fold. This average provides a robust estimate of the model's ability to generalize to unseen data, as it accounts for variations in different subsets of the dataset.

The final evaluation of the model is conducted on the testing set, and its performance is assessed using a suite of evaluation metrics. The selected metrics include Accuracy, Precision, Recall, Receiver Operating Characteristic (ROC) curve, Area Under the ROC Curve (AUC), F1-score, McNemar's test, Cost-score, and training times. Each metric serves a distinct purpose:

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}}, \quad (7)$$

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}, \quad (8)$$

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}. \quad (9)$$

Accuracy measures the overall correctness, while Precision assesses the proportion of true positive predictions among all positive predictions, and Recall evaluates the model's ability to identify all actual positive instances. In predictive maintenance, high Recall is often critical to minimize missed potential failures.

The ROC curve illustrates the trade-off between the True Positive Rate (Recall) and the False Positive Rate at different threshold settings, providing a comprehensive view of the model's discriminative ability. The AUC quantifies the overall ability to distinguish between classes, with a higher AUC indicating better performance irrespective of the chosen threshold. The F1-score, the harmonic mean of Precision and Recall, offers a balanced measure that accounts for both false positives and false negatives, making it valuable in imbalanced datasets often encountered in predictive maintenance.

McNemar's test statistically compares the performance of different classification models on the same dataset, evaluating whether there is a significant difference in the number of misclassifications between two models. This test indicates whether one model significantly outperforms another.

The Cost-score is a domain-specific metric that quantifies the economic impact of the model's predictions. It accounts for the costs associated with true positives, false positives, and false negatives, reflecting the real-world financial implications of maintenance decisions. The Cost-score is computed using:

$$C = TP \times TP_c + FP \times FP_c + FN \times FN_c, \quad (10)$$

$$C_{min} = (TP + FN) \times TP_c, \quad (11)$$

$$C_{base} = (TP + FN) \times FN_c, \quad (12)$$

$$S(C) = \frac{(C - C_{min})}{(C_{base} - C_{min})}. \quad (13)$$

Here, TP_c , FP_c , and FN_c represent the costs associated with true positives, false positives, and false negatives, respectively. The Cost-score $S(C)$ normalizes the total cost C between 0 and 1, where a score closer to 0 indicates lower costs and a score above 1 signifies higher costs compared to not performing predictive maintenance.

The total processing time for each model includes training time and prediction computation time, which refers to the time taken to analyze a single dataset sample and generate a prediction. Monitoring training times is essential for evaluating the model's scalability and efficiency in real-time industrial settings.

Hyperparameter Tuning and Optimization: Grid search within the cross-validation framework explores a range of hyperparameter values. The parameters tuned include the number of LSTM/BiLSTM layers, hidden units per layer, dropout rates, learning rates, and batch sizes. For example, the number of hidden units per LSTM layer was tested between 32 and 256, dropout rates between 0.1 and 0.5, and learning rates from 10^{-4} to 10^{-2} . The optimal hyperparameters were selected based on the highest average AUC across the validation folds, ensuring an effective balance between sensitivity and specificity.

Cross-Validation Implementation: We chose five-fold cross-validation to balance computational efficiency and reliability of performance estimates. In each iteration, the dataset is partitioned into five subsets with equal class distributions, ensuring that each fold is representative of the overall dataset. This stratification is critical for maintaining the integrity of class proportions, especially in imbalanced datasets.

Evaluation Metrics Justification: Each evaluation metric addresses a specific aspect of model performance for predictive maintenance. Accuracy provides an overall measure, but it can be misleading in imbalanced datasets. Precision and Recall offer deeper insight, with Recall particularly important to avoid missing potential failures. The ROC and AUC metrics evaluate the model's ability to separate classes across multiple thresholds. The F1-score balances Precision and Recall, while McNemar's test statistically validates differences between models. Finally, the Cost-score offers practical insights into the economic implications of misclassifications, aligning model evaluation with real-world maintenance decisions.

Comprehensive Analysis: The experimental results include a comparative performance analysis between the proposed LSTM/BiLSTM models and baseline models, such as simpler RNNs and CNN-based approaches. Sensitivity analysis is carried out to examine the impact of different hyperparameter settings on model performance. A discussion on the trade-offs between various models highlights strengths and limitations in the context of predictive maintenance, ensuring that the chosen model not only performs well quantitatively but also aligns with practical considerations.

IV. EXPERIMENTS

The results were obtained following the methodology detailed in Section III. All experiments were conducted on a Jupyter-Lab environment hosted on a server with an Intel Xeon Gold 6152 CPU @ 2.10 GHz (x86_64 architecture).

A. PROBLEM AND DATA UNDERSTANDING APPLICATION

In collaboration with engineers, maintenance personnel, and other domain experts, we identified factors most relevant to turbocharger failure modes. These include exhaust gas temperature (before and after turbocharging), air restriction and quality, pressure, engine speed and load, fuel and engine oil consumption, and oil pressure and temperature. Engaging stakeholders early ensured that the final set of variables reflected real-world operational needs.

The selected variables have been anonymized to protect proprietary information. Diagnostic Trouble Codes (DTCs) were omitted, preventing any direct association with the data-supplying company. Each variable's description remains sufficiently general for applicability across different vehicle models and manufacturers. In total, 38 candidate variables (retrievable from the vehicle's ECU) were chosen to detect and predict turbocharger failures. These include accelerator pedal position, air filter flow restriction, exhaust temperature upstream of the catalyst, engine speed, coolant temperature, and oil pressure (minimum/maximum readings), among others.

We adopted a sliding-box approach to classify whether a turbocharger is likely to fail within a specific horizon. In this binary classification scheme, component failures within a 90-day window are labeled as defective (1), while the absence of a failure record indicates a non-defective state (0). Domain experts recommended 90 days as a balanced interval for early detection of wear patterns while ensuring sufficient operational data availability.

For this research, data was assembled from 95 vehicles with documented turbocharger failures and 100 vehicles without failures, yielding 38 variables in various formats (histograms, vector matrices, and scalars). As illustrated in Fig. 6, the dataset contains a significant proportion of missing values for certain variables (e.g., ambient temperature, atmospheric pressure, fuel filter restriction). Such missingness can degrade reliability and must be addressed during preprocessing.

Moreover, only 16% of the data belongs to the positive (failure) class, indicating an inherent imbalance. Fig. 7 presents a heatmap of feature correlations: lighter colors reflect stronger correlations, while darker colors represent weaker associations. These insights highlight the importance of cleaning, imputing missing values, and employing imbalance-correction techniques.

Therefore, these preliminary analyses—conducted with maintenance and engineering stakeholders—indicate the necessity of robust imputation and class-balancing measures before applying deep learning models. Establishing this foundation paves the way for an effective predictive maintenance solution, as the subsequent methodology accounts for data

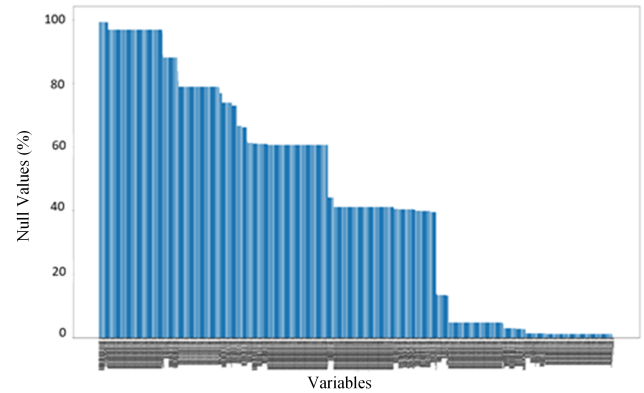


FIGURE 6. Percentage of null data.

gaps, label imbalance, and other practical constraints in real-world fleet operations.

B. DATA PREPARATION APPLICATION

Addressing the pronounced imbalance between non-defective and defective classes was essential to ensure the model could learn effectively from both categories. To mitigate this issue, we employed SMOTE, which is effective for high-dimensional, class-imbalanced data [31]. SMOTE generates synthetic samples in feature space for the minority class, expanding its representation and reducing bias in classification outcomes. Its pseudo-code is detailed by [15]; in essence, it linearly interpolates between two similar minority examples $\mathbf{X}$ and $\mathbf{X}^R$.

Subsequently, we thoroughly checked for null entries and excluded rows with high proportions of missing data. For the remaining variables with fewer null values, we used Mean Imputation (MI). MI estimates the mean value for each feature from the training data and replaces null values accordingly, chosen for its simplicity and effectiveness in preserving overall data distribution. Applying MI to the training and validation datasets ensures consistency in data preprocessing.

With SMOTE-generated samples constituting roughly 50% of the training set, the distribution of defective and non-defective examples became more balanced. This procedure mitigates the risk that the model would learn predominantly from the majority class, thereby improving its ability to detect minority class instances. Finally, we performed MinMax scaling for normalization, ensuring uniform feature ranges that avoid skewing the learning process.

These data preparation steps—missing-value management, targeted class-balancing, and normalization—form the foundation for more robust model training. They help maintain the dataset's statistical integrity and allow the deep learning algorithms to capture indicative patterns of turbocharger failures more effectively.

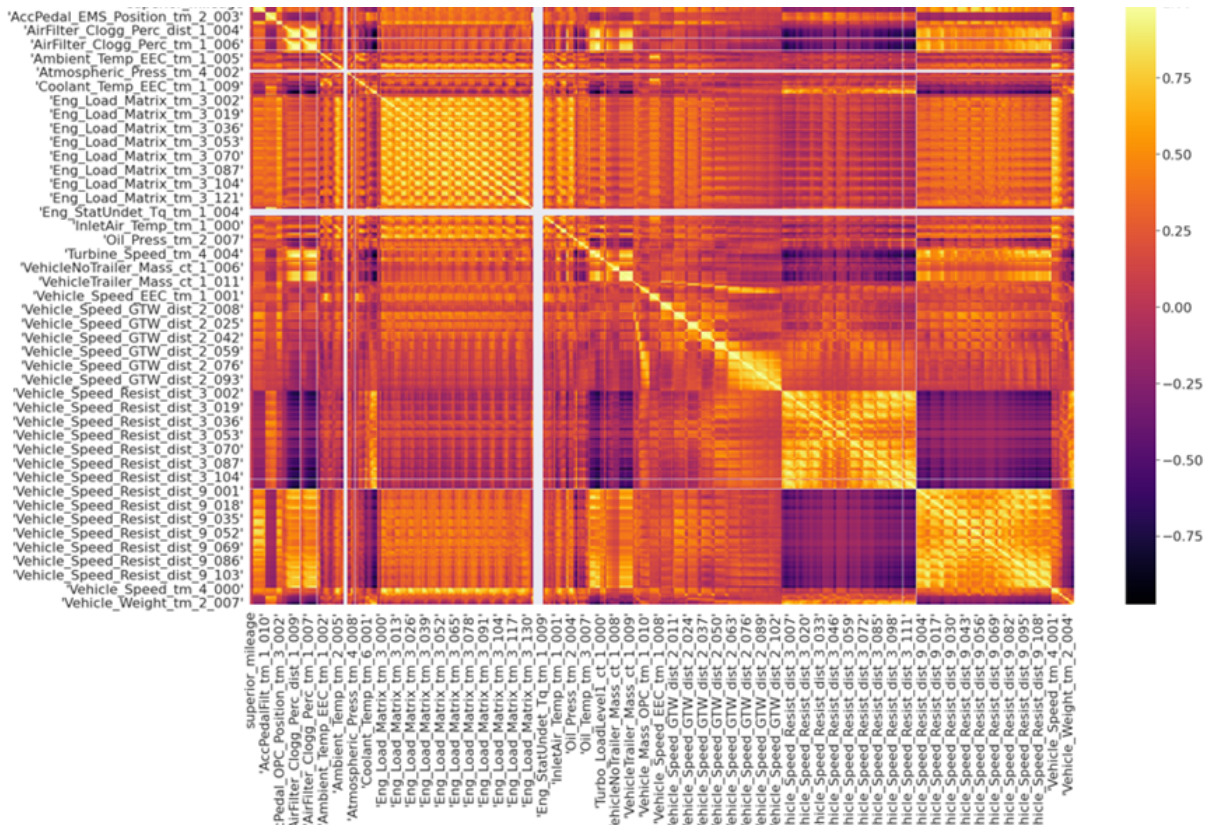


FIGURE 7. Correlation matrix of features from the commercial-vehicle dataset.

C. DEEP LEARNING MODELS APPLICATION AND EVALUATION

The architecture and hyperparameter configurations of the models, sampling techniques, and specific parameters are critical for understanding how each model processes the selected variables. LSTM and BiLSTM architectures were finalized after preliminary comparisons with simpler RNNs and CNN-based models, which exhibited lower stability and reduced accuracy under partial sensor coverage.

Fig. 8 shows the LSTM and BiLSTM architectures. The BiLSTM model adds a backward pass, capturing additional temporal cues.

Table 1 summarizes the main hyperparameters. We explored simpler RNNs and CNN-based approaches in pilot tests, but LSTM/BiLSTM was more stable and accurate for our dataset size and partial sensor coverage. An epoch value of 400 was selected after verifying convergence curves.

TABLE 1. Parameter settings for LSTM and BiLSTM.

Hyperparameter	Configuration
Optimizer	Adam (Learning rate = 0.001)
Batch size	32
Epochs	400
Sampling	SMOTE
Number of Neighbors (SMOTE)	5
Number of Folds (CV)	5

Five-fold cross-validation was applied to avoid overfitting and improve generalization. SMOTE [15] was used before each fold split to maintain realistic class balance. Mean imputation addressed missing values, and sensitivity analyses revealed negligible performance shifts when toggling SMOTE.

Fig. 9 shows the accuracy and loss curves for both LSTM and BiLSTM models across epochs. Both networks converged steadily, reflecting consistent learning:

Figs. 10 and 11 display the confusion matrices. Although both networks achieve robust classification, BiLSTM reduces the number of false negatives to 17, a critical advantage in predictive maintenance.

Table 2 presents the evaluation metrics, including Accuracy, Precision, Recall, and F1-score, along with a cost-based metric (Cost-score). While LSTM has a marginally higher Precision, BiLSTM exhibits stronger Recall (98.65%) and a lower cost-score of 0.01. A simpler baseline RNN model (not shown) achieved around 88% accuracy, underscoring LSTM/BiLSTM's performance edge.

Figs. 12 and 13 provide ROC curves with $AUC \approx 0.993$. This near-perfect discrimination aligns with the confusion matrix findings.

Table 3 presents McNemar's test:

Since $p < 0.05$, the null hypothesis is rejected, indicating that the two models perform significantly differently. BiLSTM's higher recall may be particularly suitable for predictive maintenance scenarios where missed failures are espe-

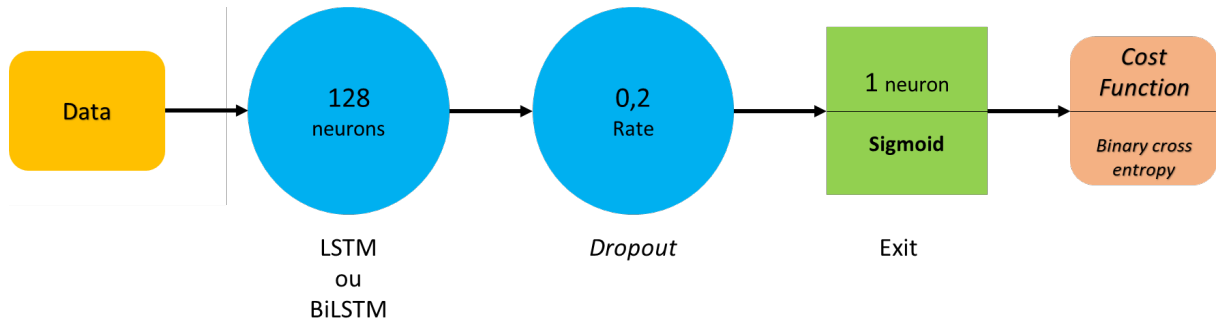


FIGURE 8. The architecture of LSTM and BiLSTM. BiLSTM incorporates backward and forward passes.

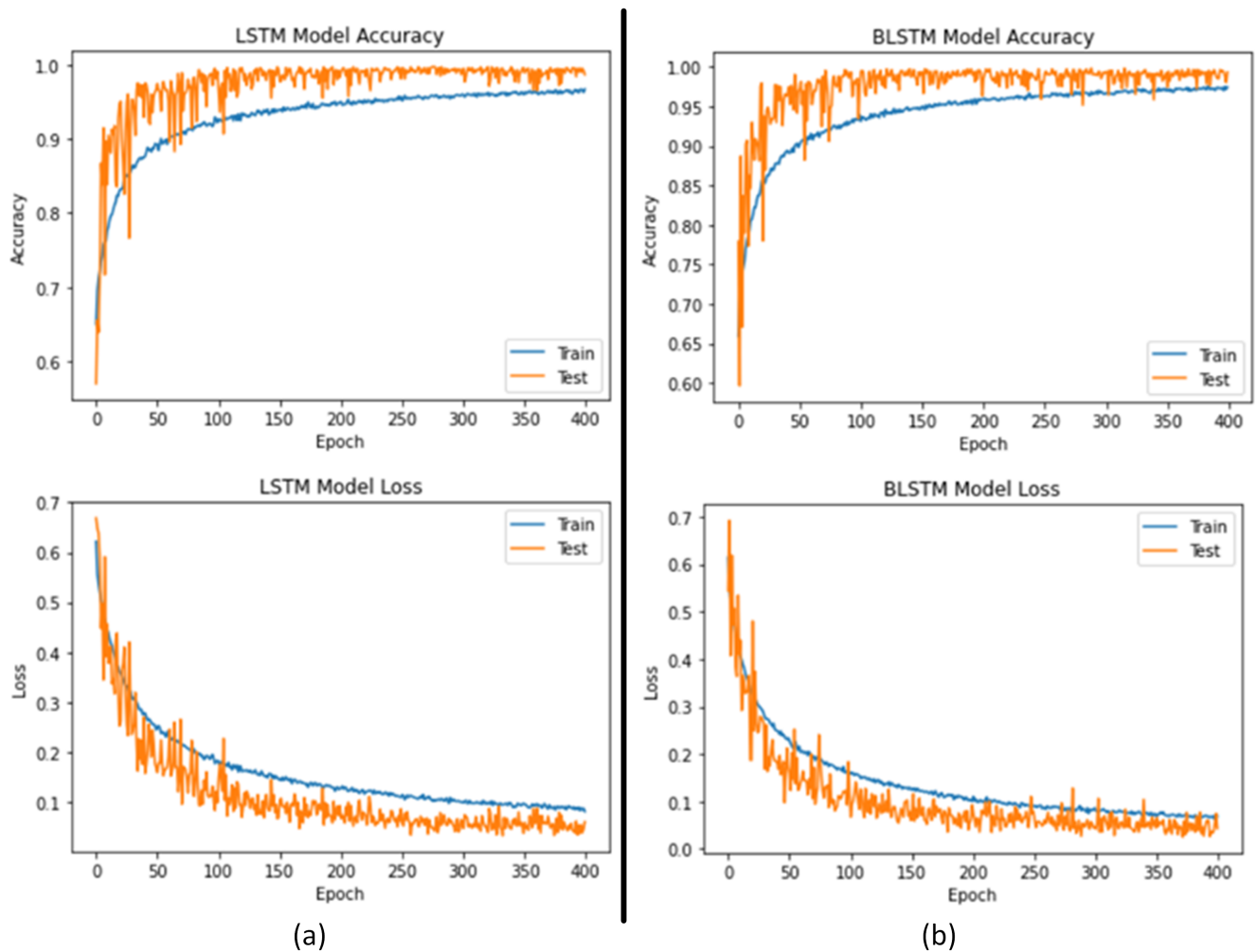


FIGURE 9. Accuracy (top) and loss (bottom) in training the LSTM (a) and BiLSTM (b) models.

TABLE 2. Evaluation metrics for the LSTM and BiLSTM models (from Eq.7 to Eq.13).

Model	Accuracy	Precision	Recall	F1-score	Cost-Score	Training Time (s)	Forecast Time (s)	Total Time (s)
LSTM	96.63%	95.72%	97.62%	96.66%	0.02	451.47	0.55	452.02
BiLSTM	96.87%	95.25%	98.65%	96.92%	0.01	550.93	0.59	551.52

cially costly.

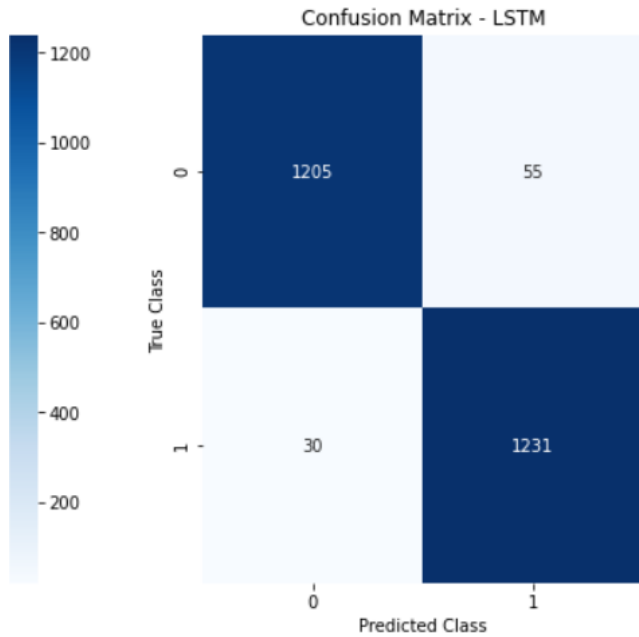


FIGURE 10. Confusion matrix for the LSTM network.

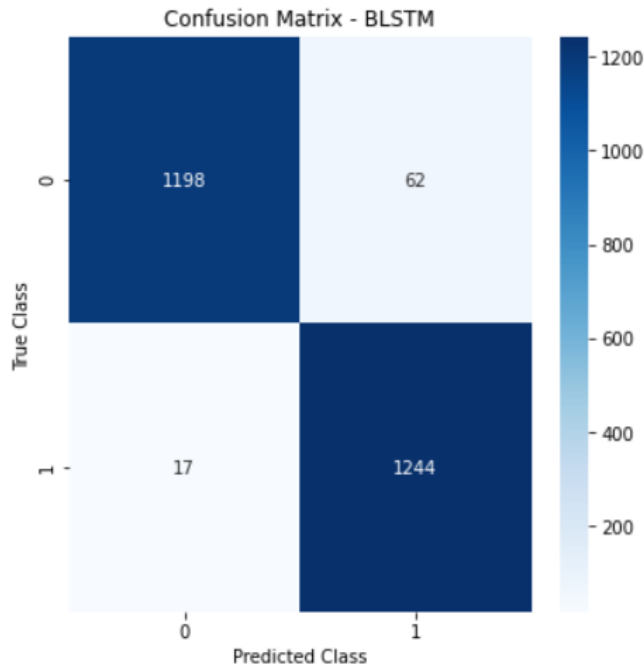


FIGURE 11. Confusion matrix for the BiLSTM network.

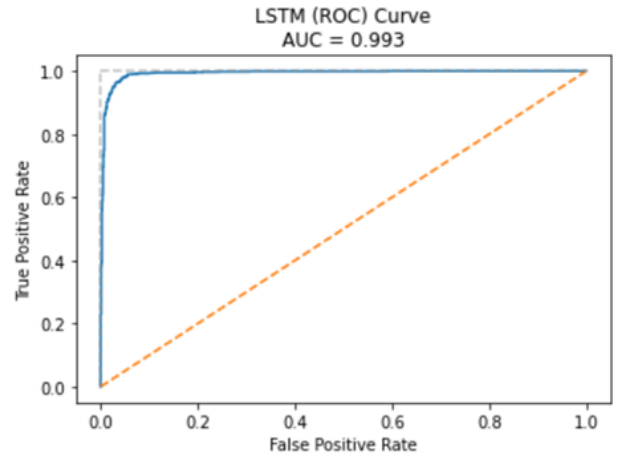


FIGURE 12. LSTM ROC curve.

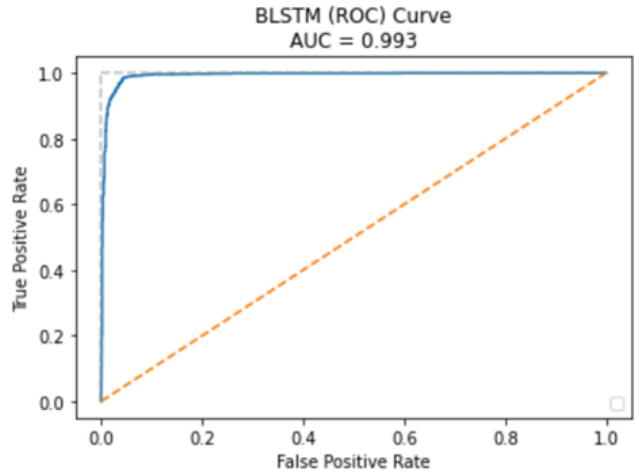


FIGURE 13. BiLSTM ROC curve.

TABLE 3. McNemar's test result comparing LSTM and BiLSTM.

Comparison	p-value	Null hypothesis
LSTM vs. BiLSTM	0.0092	Rejected

D. RESULTS ANALYSIS

BiLSTM's superior recall (98.65%) makes it well-suited for predictive maintenance, where missing an actual failure (false negative) can be highly detrimental. Although LSTM exhibits a slightly higher precision (95.72%), the overall F1-score favors BiLSTM. Workshop stakeholders often prefer fewer missed failures at the expense of handling some additional false positives.

However, BiLSTM requires more computational overhead in training and prediction, which could be a concern if rapid, on-the-fly diagnostics are needed. Future research could explore pruning strategies or lightweight Transformer variants to reduce BiLSTM's runtime while maintaining performance.

Data preprocessing notably influenced outcomes. Mean

imputation did not significantly degrade model accuracy, and SMOTE balanced the minority class without inflating results artificially. Testing without SMOTE led to a 4–5% drop in recall for LSTM and BiLSTM, underscoring the importance of synthetic oversampling in this highly imbalanced setting.

The cost-score metric (near 0.01 for BiLSTM) highlights the real-world benefits: most critical failures are detected before leading to expensive downtime. Although these results use a single fleet dataset, they indicate that properly tuned BiLSTM architectures can provide robust early failure detection, especially when partial sensor coverage necessitates learning from multiple data streams.

V. CONCLUSION

This study proposed a methodology for diagnosing and predicting vehicle faults by integrating specification data, maintenance history, and operational parameters. The primary objective was to improve repair times, standardize data collection, and strengthen predictive maintenance strategies for mechanical systems in commercial vehicles. The resulting predictive model outperformed existing approaches by leveraging real-world industry data and data mining techniques, underscoring the practical impact of unifying multiple data sources.

The experiments demonstrated that model selection is critical for optimal performance in predictive maintenance scenarios. Applying McNemar's test revealed statistically significant differences, reinforcing the importance of architectures matched to the data characteristics and task requirements.

For future research, several directions will be explored. First, an automated methodology to streamline data acquisition, storage, and organization will be developed to facilitate maintenance analyses on a broader scale. Second, more advanced preprocessing and hyperparameter tuning techniques will be examined to further refine model performance. Finally, alternative neural network architectures, including transformer-based models, will be explored to assess their utility in predictive maintenance for various mechanical systems and broader industrial contexts.

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